



Title

From CHI Paper Review to ICCA 2026 Conference Paper

Explainable AI-Based Educational Chatbot for
Personalized Learning Support

Name: Naznin Akter Roshmny

ID: 22-46092-1 ; Section: B

Course Name: Human-Computer Interaction (HCI)

Course Teacher: Prof. Dr. Tabin Hasan

Paper Overview

Original Research Area

Human-Computer Interaction (HCI)

Research Problem

- Traditional AI educational chatbots provide answers but often fail to explain how answers are generated. This reduces user trust, learning effectiveness, and interaction quality.

The proposed system aims to improve trust, usability, and learning experience through Explainable AI.

Motivation

- Students need:
 - Transparent AI responses
 - Better learning interaction
 - Personalized educational support
 - User-friendly educational systems

METHOD SUMMARY

Original System Methodology

Core System Idea

The proposed educational chatbot uses:

- Natural Language Processing (NLP)
- Explainable AI (XAI)
- Personalized recommendation systems
- Adaptive user interfaces

The original chatbot system:

- Student asks question
- NLP processes query
- AI generates response
- Explainable AI module creates explanation
- Personalized recommendation generated
- Student receives adaptive feedback

Workflow:

Student → Chatbot → AI Processing → Generated Response

Technologies & Dataset

❑ Technologies Used

- Artificial Intelligence
- Machine Learning
- NLP
- Explainable AI
- HCI Design Principles

❑ Possible Dataset Sources

- Educational QA datasets
- Student interaction logs
- Learning analytics data

❑ Interface Features

Interactive chatbot UI

Adaptive feedback system

Accessibility support

KEY CONTRIBUTIONS

❑ Contributions of Original Work

- Developed AI-based educational chatbot
- Improved student interaction
- Automated learning assistance
- Provided intelligent educational support
- Increased student engagement

Limitations & Research Gaps

Major Limitations

1. Lack of Explainability

Students cannot understand:

- why the AI generated the answer
- reasoning process of the chatbot

2. Limited User Trust

- Users may not trust AI-generated educational responses.

3. Small User Study

- The system was tested on limited participants.

4. Weak Accessibility

No strong support for:

- visually impaired users
- diverse learning needs

More Research Gaps

Additional Problems

No Personalized Learning

- All users receive similar interaction.

Weak Evaluation Metrics

- Only system accuracy measured. Missing:
- usability analysis
- UX evaluation
- trust measurement

Scalability Problems

- System not tested in real-world large-scale educational environments.

Ethical Concerns

- Privacy and AI transparency not deeply discussed.

Proposed Improvements

Proposed Enhanced HCI Framework

The improved system introduces:

Explainable AI Layer

The chatbot explains:

- answer generation process
- reasoning steps
- confidence level

Personalized Learning Engine

System adapts based on:

- student performance
- learning speed
- user preferences

Adaptive User Interface

Different UI modes for:

- beginners
- advanced learners
- disabled users

Advanced Technology Integration

AI/ML Features

- Explainable AI
- Recommendation systems
- Adaptive learning algorithms

HCI Features

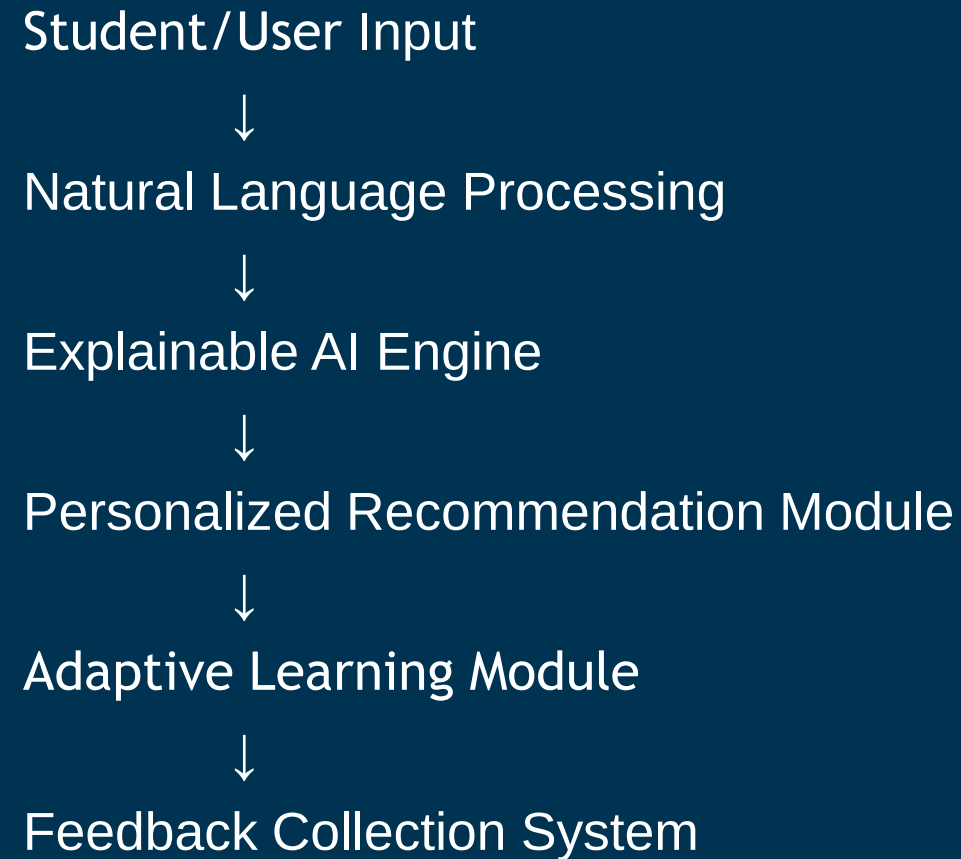
- User-centered design
- Accessibility support
- Better interaction flow

Future Technologies

- AR/VR learning support
- Voice-based interaction
- Emotion-aware educational assistant

SYSTEM ARCHITECTURE

Proposed Architecture



EVALUATION PLAN

Participants:

40 university students

Tasks:

- Ask educational questions
- Evaluate explanations
- Solve learning activities
- Provide usability feedback

Evaluation Metrics:

- Usability Score (SUS)
- User Satisfaction
- Trust Level
- Task Completion Time

BASELINE COMPARISON

System Comparison

Feature	Original System	Proposed System
Explainable AI	No	Yes
Voice Support	No	Yes
Accessibility	Limited	Improved
Personalization	No	Yes
User Trust	Medium	High

EXPECTED OUTCOMES

The proposed system is expected to:

- improve student trust
- increase usability
- reduce cognitive load
- support accessibility
- enhance learning interaction

CONCLUSION

- ACM CHI Conference Papers
- Human-Computer Interaction Research
- Explainable AI Literature
- Educational Chatbot Systems
- AI-assisted Learning Research

REFERENCES

- ACM CHI Conference Papers
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- AI-assisted Learning Research



Thank You